

MMALS G2: A Geometric Perspective on Continual Functional Memory

From Chronicle and inferred-context routing to admissible adaptation flows

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Abstract

This paper proposes the G2 geometric perspective for Mycelium-Inspired Mutualistic Adaptive Learning Systems (MMALS). It starts from the evidence discipline established in Geometry-MMALS G1, the route-first control model of RC2O/TPUT, the dual reconstructive-compiled memory of CHRONICLE, and the safety role of MMALS-CAL. It interprets continual learning not merely as sequential optimization, but as controlled transport of functions, routes, memories, and audit distributions across coupled geometric spaces. The selected paper by Perez-Dattari et al. on Stable Flow Matching Dynamical Systems supplies a decisive engineering pattern: expressive learned dynamics should be restricted to admissible velocity sets and stabilized either softly or by construction. G2 adapts that pattern cautiously. It does not assert that learning trajectories are physical trajectories, nor that a valid Lyapunov function already exists for arbitrary continual adaptation. Instead, it defines falsifiable geometric objects, candidate invariants, admissible transition cones, route-function alignment measures, and an experiment programme for testing whether geometric control adds value beyond non-geometric baselines under matched budgets.

Keywords: continual learning, inferred context, functional memory, routing geometry, information geometry, Lyapunov stability, flow matching, conformal prediction, engineering assurance.

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1 Research boundary and central thesis

G2 is not the claim that neural learning literally follows a physical manifold, nor that every adaptation can be assigned a conserved energy. It is the narrower proposition that the operational state of a continual adaptive system can be represented by geometric objects whose distances, tangent directions, transport maps, and admissible regions predict or improve behaviour under matched budgets.

A continual learner becomes engineerable when context inference, route selection, functional memory, adaptation, and safety calibration are treated as coupled but non-identical geometric processes, and when adaptation is restricted to transitions that are empirically admissible for the current objective, risk envelope, and evidence state.

This follows the G1 evidence discipline [2]: descriptive order is insufficient. Predictive, causal, and operational evidence must agree. A smooth embedding is not a contribution if hosts are functionally redundant, route identity is unstable across seeds, or geometric regularization merely suppresses plasticity.

G2 begins only after a claim ladder is passed: pipeline validity; reconstructable audit geometry; grounded order beyond backbone and shuffled nulls; held-out prediction; causal intervention; matched-budget operational gain; cross-benchmark robustness; and finally geometric control that outperforms simpler alternatives. Geometry is admitted because it changes an engineering decision for the better, not because it is elegant.

2 Continual learning, Chronicle, and geometry

2.1 Continual learning as historical control

Classical continual learning studies the stability–plasticity problem: learning new distributions without unacceptable loss of previous capabilities [5, 6]. The standard state variables are parameters, gradients, replay buffers, task identities, and performance matrices. MMALS adds two demands.

First, context identity must be inferred rather than assumed:

$$q_t(c \mid x_t, m_t), \quad (1)$$

where m_t is the historical memory state. Second, what must persist is not merely data or weights, but a routable capability: a transformation that can be selected, combined, audited, and judged by its marginal contribution to the ecosystem.

Continual learning therefore becomes a path-dependent control problem. At each time, the system decides whether to reuse, protect, adapt, merge, compile, retire, or allocate capability. Two systems with identical current accuracy may have different future options because their memory structure, host specialization, calibration evidence, and irreversible decisions differ.

2.2 Chronicle as typed historical memory

CHRONICLE is the MMALS historical substrate. It separates:

$$M_R : \text{reconstructive memory of evidence, routes, decisions, failures, and provenance,} \quad (2)$$

$$M_S : \text{compiled functional memory of reusable macro-routes and validity envelopes.} \quad (3)$$

M_R supports replay, causal diagnosis, and later reinterpretation. It preserves raw evidence immutably. M_S supports efficient action, but compression is admitted only when decision-relevant and audit-relevant information is retained.

A useful abstraction is a typed graph

$$\mathcal{C}_t = (V_t, E_t, \Phi_t, \Pi_t), \quad (4)$$

where vertices include contexts, hosts, routes, calibration states, memory objects, claims, and experiments; edges include temporal, causal, specialization, derivation, and invalidation relations; Φ_t maps functional states to audit traces; and Π_t defines admissible lifecycle operations.

2.3 Why geometry helps

Geometry supplies a language for four questions that scalar metrics obscure:

1. **Neighbourhood:** which contexts, routes, and functions are locally similar?
2. **Direction:** which update directions preserve useful function and which cause interference?
3. **Transport:** how can a capability move across context without arbitrary distortion?
4. **Boundary:** which transitions are admissible under retention, cost, safety, and audit constraints?

The qualification is essential: functional sharing, representational proximity, and calibration compatibility need not define the same graph.

3 The G2 multi-space model

G2 distinguishes five spaces rather than forcing one universal manifold.

Observation space $\mathcal{X}$. Raw inputs and environmental states.

Context space $\mathcal{M}_C$. Latent inferred regimes. A context is not necessarily a task label; it is an explanatory state that changes routing, calibration, or adaptation.

Route space $\mathcal{M}_R$. Routes are first-class structured objects

$$r = (c, B, a_M, T, d), \quad (5)$$

including plausible context, host coalition, memory access, transformation, and decision.

Functional quotient space $\mathcal{M}_F$. Parameters are identified when they implement the same relevant behaviour:

$$\theta_1 \sim \theta_2 \iff F_{\theta_1} \equiv F_{\theta_2} \text{ under the declared intervention and audit schema.} \quad (6)$$

The quotient $\Theta/\sim$ avoids treating reparameterization as functional change.

Audit-distribution space $\mathcal{M}_A$. The observation map

$$\Phi : F \mapsto P(A \mid x, t, g) \quad (7)$$

turns function into distributions of evidence traces. Information geometry is natural here [9, 10]. Directions in $\ker D\Phi$ are audit blind spots: behaviour changes while the evidence schema remains locally insensitive.

Calibration space $\mathcal{M}_{cal}$. Calibration states include conformity-score distributions, thresholds, local coverage errors, set sizes, and action consequences. This space must not be conflated with context space.

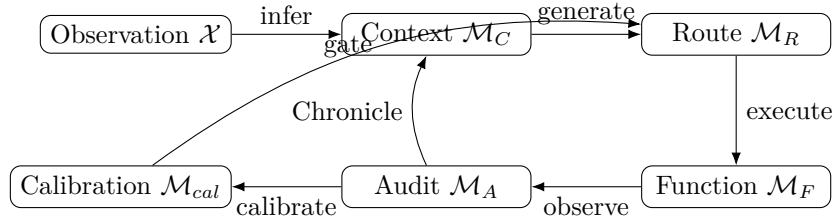


Figure 1: G2 couples multiple spaces with distinct semantics and evidence.

Each space may carry its own metric. Cross-space maps induce pullbacks. For example, $\Phi : \mathcal{M}_F \rightarrow \mathcal{M}_A$ induces $g_F^{audit} = \Phi^* g_A$, measuring only functional variation visible through the audit schema. A router map into the probability simplex can use a stable Hellinger baseline

$$d_H(p, q) = \frac{1}{\sqrt{2}} \|\sqrt{p} - \sqrt{q}\|_2. \quad (8)$$

Fisher–Rao should remain an evaluation metric until gradient stability is demonstrated.

4 Stable Flow Matching Dynamical Systems as a design pattern

Perez-Dattari et al. introduce Stable Flow Matching Dynamical Systems, combining expressive multimodal flow matching with positive invariance and Lyapunov stability constraints [1, 11]. The key design is to learn a stochastic vector field while constraining generated velocities to an admissible set. The paper offers soft penalty-based and hard structurally projected variants and extends the construction to Lie groups.

The contribution to MMALS is not the robot application or the flow-matching loss. It is the pattern:

1. expressive learned dynamics can be restricted to an admissible solution set;
2. safety constraints can be soft or architectural;
3. latent Lyapunov functions may simplify stability conditions;
4. non-Euclidean state spaces should be respected where physically or functionally justified.

The analogy is

$$\dot{x} \in \dot{\mathcal{X}}_A \iff \delta s \in T_s \mathcal{M}^{adm}, \quad (9)$$

where s is an MMALS state containing route, function, memory, calibration, cost, and audit variables. The admissible tangent cone contains update directions satisfying declared constraints.

The limits matter. SFMDS studies continuous-time trajectories with a defined equilibrium. MMALS studies historical learning systems whose objectives and coordinates can change. G2 must not claim global asymptotic stability unless a meaningful equilibrium or invariant set is defined and the conditions are proved.

A soft constraint adds penalties:

$$L_{G2} = L_{task} + \lambda_R L_{ret} + \lambda_C L_{cost} + \lambda_D L_{drift} + \lambda_A L_{audit} + \lambda_S L_{safe}. \quad (10)$$

A hard constraint projects a proposed transition v onto an admissible cone:

$$v^\Pi = \Pi_{\mathcal{T}_{adm}(s)}(v). \quad (11)$$

Candidate inequalities are

$$\Delta \text{Retention}(v) \geq -\epsilon_R, \quad (12)$$

$$\Delta \text{Risk}(v) \leq \epsilon_S, \quad (13)$$

$$\text{Cost}(v) \leq B, \quad (14)$$

$$\text{AuditVisibility}(v) \geq \eta_A. \quad (15)$$

Hard projection is justified only if these quantities can be estimated reliably and projection does not destroy learning.

5 From route selection to admissible adaptation flow

The minimal TPUTstate is [3]

$$s_t = (z_t, \tilde{q}_t(c), m_t, g_t, D_t, B_t), \quad (16)$$

and routes are scored with named components:

$$E(r) = E_{fit} + E_{mem} + E_{ret} + E_{cost} + E_{drift} + E_{spec} + E_{audit} + E_{safe}. \quad (17)$$

G2 changes the role of energy. It is not only a ranking score; its local derivatives can define candidate directions while constraints define admissibility.

5.1 Route–function alignment

A router may stay smooth while host function drifts. Define

$$A_{r,h}(t) = \text{align}(g_{route,h}(t), \mathcal{S}_{func,h}(t)), \quad (18)$$

where $g_{route,h}$ is a local route preference direction and $\mathcal{S}_{func,h}$ is estimated from functional interventions or Jacobians. A fall in alignment produces a Chronicle event: route semantics drift. Responses include recalibration, router repair, host repair, retirement, or abstention.

5.2 Counterfactual route regret

For every admissible route, evaluate frozen execution:

$$R_{route}(x) = L(r_{selected}, x) - \min_{r \in \mathcal{R}_{adm}(x)} L(r, x). \quad (19)$$

This separates host incapacity from router misallocation. Full counterfactual search is primarily an offline evaluator unless runtime cost is acceptable.

5.3 Capacity and saturation

New capacity should be allocated only when evidence separates representational saturation from plasticity saturation. A host creation action is admissible when existing routes have high counterfactual regret, adaptation directions are blocked by justified constraints, and robust calibration rejects current capability. The allocation must be compared with replay, EWC, progressive architectures, and sparse MoE baselines [5, 7, 8].

6 Chronicle as a geometric historical system

Every reconstructive record should contain

$$A_t = (x_t, z_t, \tilde{q}_t, R_t, E_t, \pi_t, r_t^*, y_t, g_t, \Gamma_t, \text{cost}_t, \text{version}_t). \quad (20)$$

A compiled memory object is a macro-route plus a validity envelope

$$\mu = (r_{\text{macro}}, \Omega_C, \Omega_F, \Omega_{\text{cal}}, B_{\text{max}}, \tau_{\text{expiry}}, \mathcal{P}). \quad (21)$$

The envelope is not merely a ball in latent space. It is the intersection of context, functional, calibration, cost, and provenance conditions.

A transition between capabilities must be evaluated in three relations:

$$G_{\text{geom}}(i, j) : \text{representational proximity}, \quad (22)$$

$$G_{\text{func}}(i, j) : \text{functional substitutability or positive transfer}, \quad (23)$$

$$G_{\text{cal}}(i, j) : \text{calibration compatibility}. \quad (24)$$

A merge is admissible only when the relevant relations support it. Proximity alone is insufficient.

Retirement becomes a value decision:

$$\text{retain}(h) \iff \mathbb{E}[V_{\text{reuse}}(h)] > C_{\text{capacity}}(h) + C_{\text{interference}}(h) + C_{\text{audit}}(h). \quad (25)$$

The uncertainty of this estimate and any irreversible loss must be recorded.

7 Safe conformal calibration as a geometric gate

Conformal prediction supplies finite-sample marginal coverage under stated assumptions [12, 13]. Adaptive conformal inference can respond to shift [14]. Neither is a system safety proof. MMALS-CAL is one gate in a layered assurance chain.

For high-confidence context inference,

$$\Gamma_c(x) = \{y : S_c(x, y) \leq q_{1-\alpha, c}\}. \quad (26)$$

For ambiguous context inference, preserve several plausible contexts and construct a validated robust aggregation

$$\Gamma_{\text{rob}}(x) = \mathcal{R}(\{\Gamma_c(x) : c \in \text{TopK}(q)\}). \quad (27)$$

This implements non-premature commitment.

CAL should report coverage error, set size, calibration inflation, prediction bias, action and abstention rates, and severity-weighted violations. A calibrator can hide routing failure by widening sets. Chronicle should therefore separate retention failure, inference failure, functional drift, and calibration compensation.

A host may have a calibrated latent acceptance region

$$\mathcal{C}_h = \{z : g_h(z) \leq q_{1-\alpha,h}\}. \quad (28)$$

An input outside all regions triggers abstention, robust multi-host evaluation, or capacity investigation. Geometric inclusion does not imply functional correctness; host-specific outcomes and interventions remain mandatory.

8 Candidate Lyapunov perspectives

A Lyapunov-like candidate would measure distance from a desired operational set, not necessarily a point equilibrium:

$$\mathcal{S}_{safe} = \{s : \text{risk} \leq \rho, \text{retention} \geq R_{min}, \text{cost} \leq B, \text{auditability} \geq A_{min}\}. \quad (29)$$

A set-stability candidate is $V(s) = d(s, \mathcal{S}_{safe})^2$. An admissible direction may satisfy

$$\nabla V(s)^\top v \leq -\kappa V(s) + \xi_t, \quad (30)$$

where ξ_t permits controlled adaptation under non-stationarity.

Strict monotonic decrease may be wrong: useful learning can temporarily increase loss, uncertainty, or cost. G2 should examine hard invariance for safety constraints, bounded regret, recovery to acceptable sets, input-to-state stability with drift as disturbance, and multiple Lyapunov functions for different regimes.

The framing is falsified when the candidate does not predict failures, adds no utility, creates severe intransigence, or depends arbitrarily on representation coordinates.

9 Lie groups, subspaces, Koopman, and mechanics

SFMDs use Lie groups because robot orientations genuinely live on $SO(3)$ or $SE(3)$. MMALS should use Lie structure only where a real compositional symmetry exists: invertible route transformations, operator-valued histories, or physical pose applications. A host bank is not automatically a group; closure and inverses may fail.

If context or function is subspace-valued, routing may be studied on a Grassmann manifold. This is admitted only if subspace routing outperforms vector prototypes under equal compute and improves causal intervention specificity.

Koopman methods suggest that nonlinear adaptation trajectories may become simpler in a lifted observable space. Euler–Poincaré mechanics suggests constrained evolution under symmetry reduction. Both are mathematical foundations, not implementation mandates. Their value must be measured by prediction, control, or audit of real adaptation trajectories.

10 Engineering architecture

A deployable G2 system separates deterministic infrastructure from learned components:

1. context inference service producing sparse $q(c \mid x, m)$ and uncertainty;
2. route generator enumerating feasible host and memory coalitions;

3. energy evaluator computing named objective and constraint terms;
4. admissibility projector rejecting or projecting unsafe adaptation actions;
5. host executor running selected specialists;
6. CAL service producing sets, diagnostics, and action gates;
7. Chronicle as immutable evidence log and compiled-memory registry;
8. counterfactual auditor for offline route and host interventions;
9. lifecycle manager for create, reuse, merge, repurpose, and retire.

Algorithm 1 Minimal G2 route and adaptation loop

Require: input x , Chronicle m , goal g , budget B

```

1:  $z \leftarrow P_\phi(x)$ ;  $q \leftarrow \text{ContextInferer}(z, m)$ 
2:  $R \leftarrow \text{GenerateRoutes}(\text{TopK}(q), m, g, B)$ 
3: for  $r \in R$  do
4:   compute energy  $E(r)$ , constraints  $C(r)$ , and calibration state  $\Gamma(r, x)$ 
5: end for
6:  $R_{adm} \leftarrow \{r : C(r) \leq 0 \wedge \Gamma(r, x) \text{ passes}\}$ 
7:  $r^* \leftarrow \text{SimplestAdequateRoute}(R_{adm}, E, g, B)$ 
8:  $y \leftarrow \text{Execute}(r^*, x, m)$ 
9: append complete audit record to  $M_R$ 
10: if compilation gate passes then update  $M_S$  with macro-route and validity envelope
11: end if
12: propose lifecycle/update direction  $v$  and apply  $\Pi_{\mathcal{T}_{adm}}(v)$ 
13: record rejected and projected actions
14: return  $y$ 

```

Every output must be reproducible from model version, route candidates, energy decomposition, constraint checks, calibration state, memory reads, and lifecycle writes. An evidence store maps claims to runs, seeds, datasets, and code commits.

11 Experimental programme and falsification

11.1 G2-0: instrumentation before architecture

Use the current RC2O/MMALS host bank. Add counterfactual route regret, route–function alignment drift, context-specific versus robust CAL, calibration inflation after misrouting, and saturation signals before allocation.

11.2 G2-1: soft admissibility

Add penalty-based retention, risk, cost, and audit constraints. Compare against the same architecture without geometry and against validation-selected continual-learning baselines.

11.3 G2-2: hard projection

Project route or lifecycle actions into a validated admissible region. Report rejected actions, projection magnitude, accuracy, forgetting, intransigence, latency, and compute.

11.4 G2-3: geometric transport

Test learned or analytic transport of capabilities between nearby contexts. Include Euclidean and shuffled-geometry nulls.

11.5 G2-4: manifold-valued routing

Only if vector prototypes fail, test subspace or manifold route objects. Require matched-cost gain and cross-seed functional identity.

Endpoint	Required evidence
Retention/plasticity	Final average accuracy, forgetting, transfer, intransigence.
Context inference	Accuracy where labels exist, route regret, uncertainty, novelty detection.
Specialization	Ablation locality, intervention specificity, cross-seed function matching.
Geometry	Held-out interpolation, null comparisons, operational gain.
CAL safety	Coverage, local undercoverage, set size, inflation, severity-weighted violations.
Chronicle utility	Reconstruction fidelity, compilation speedup, envelope failures.
Cost	Training and active FLOPs, latency, memory, energy proxy.
Auditability	Complete traces, claim manifest, replayability, blind-spot analysis.

Core ablations remove route geometry, transport, functional distances, context-conditioned CAL, robust fallback, or audit constraints; freeze router or hosts separately; shuffle context geometry; and compare hard projection with matched-strength soft penalties.

The anti-metaphor safeguards are: no geometry without invariance tests; no stability claim without plasticity; no calibration claim that hides widening; no hidden task label renamed as context; no learned auditor accepted without independent validation; no oracle counterfactual presented as runtime; no Lie or Grassmann machinery without baseline failure; and no irreversible lifecycle action without provenance.

12 Engineering view

From an engineering standpoint, G2 is a control and assurance layer around an adaptive system. The primary object is not the manifold; it is the decision contract. Inputs, model state, memory state, objective, and resource budget lead to candidate routes. Deterministic checks and learned estimates classify candidates as admissible or inadmissible. The selected route is executed, calibrated, and audited. Any adaptation is treated as a change request with evidence, impact analysis, and rollback semantics.

The architecture should be modular and nearly decomposable in the sense of Simon [15]. Learned models can propose context, route, and value estimates, but evidence schemas, versioning, hard safety rules, and replay must remain deterministic where possible. Geometry becomes useful when it provides stable interfaces: distances with declared invariances, projections with measurable residuals, and transport maps with validity envelopes.

For deployment, the main engineering artefacts are:

- a route schema and energy decomposition;

- a typed Chronicle event model;
- a constraint and projection API;
- a calibration evidence contract;
- an experiment and claim manifest;
- observability traces connecting specification to evidence;
- a lifecycle state machine for host and memory objects.

13 A twelve-year-old view

Imagine a school with several small expert teams. One team understands rotated pictures, another colours, another remembers old rules. A coordinator chooses who should help.

Chronicle is the school's history book. It does not only keep old worksheets. It records which team helped, what was decided, what failed, and which shortcuts are safe to reuse.

Geometry means drawing several maps of the school. One map shows which teams understand similar things. Another shows which teams are safe for a problem. Another shows which shortcuts can be reused. The maps are not always the same.

The SFMDS paper is about a robot that can invent many movements but is not allowed to choose dangerous directions. MMALS G2 borrows the idea: the AI may learn in many ways, but some updates are blocked because they would erase important skills, cost too much, or create an unsafe answer.

A soft rule says, "That direction is bad; pay a penalty." A hard rule says, "That direction is outside the safe zone; you cannot take it." Hard rules are stronger, but they may also stop useful learning, so engineers test both.

Conformal calibration is like the coordinator saying, "I am not sure enough, so I will give several possible answers or ask for help." It is not magic safety. It helps the system know when to slow down, use several teams, or abstain.

The scientific goal is not a beautiful map. The map matters only if the school remembers, learns, saves effort, and avoids bad decisions better than a simpler school.

14 Reference articles explained through both views

14.1 Stable Flow Matching Dynamical Systems

Engineering view. Learn an expressive vector field, derive admissible velocity sets from invariance or Lyapunov conditions, and enforce them by penalty or structural projection. For MMALS, this is a design pattern for an admissibility layer around routes and lifecycle actions.

Twelve-year-old view. The robot can invent many paths, but a safety fence removes paths that could make it crash. MMALS can put a similar fence around learning steps that would erase useful knowledge.

14.2 Geometry-MMALS G1

Engineering view. G1 is the qualification programme: descriptive, predictive, causal, and operational evidence. Route smoothness without function is rejected.

Twelve-year-old view. A colourful map does not prove the teams are different. We switch teams off and swap them to test whether each one really has a special job.

14.3 MMALS-TPUT and RC2O

Engineering view. TPUT makes the route a first-class object conditioned on context, memory, goals, and budget. Named energy terms and traces provide transparent control before reinforcement learning.

Twelve-year-old view. The coordinator does not always choose the fastest team. It also thinks about memory, cost, safety, and what will be useful tomorrow.

14.4 The structural prototype

Engineering view. Frozen paths provide a structural no-forgetting mechanism under strong assumptions. This proves one memory mechanism, not the complete MMALS ecosystem [4].

Twelve-year-old view. Some notebooks are locked so new writing cannot erase them. That protects old lessons, but too many locked notebooks can make learning slow and expensive.

14.5 Information geometry

Engineering view. Probability distributions form a manifold. MMALS initially uses this to measure route and audit-distribution change, with stable computational baselines before advanced metrics.

Twelve-year-old view. Two answer patterns can be compared by all their probabilities, not only by the top answer.

14.6 Conformal prediction

Engineering view. Conformal methods turn scores into prediction sets with calibrated marginal error under explicit assumptions. Under drift, thresholds and local diagnostics must adapt.

Twelve-year-old view. When the AI is unsure, it gives a safe list instead of pretending to know one answer. We still check that the list does not become huge or miss important cases.

14.7 Mixture of experts and continual learning

Engineering view. MoE supplies sparse capacity and routing; continual learning supplies stability–plasticity benchmarks. MMALS’s possible originality is functional memory, route history, ecological contribution, cost, and evidence semantics.

Twelve-year-old view. Having many teams is not new. The new part is keeping a trustworthy history of what each team can do, what it costs, and when it is safe to reuse.

15 PhD contribution and roadmap

The doctoral contribution is a theory–engineering programme with four linked outputs:

1. a multi-space geometric formalization of inferred-context continual functional memory;
2. an admissible-adaptation framework connecting route energy, Chronicle lifecycle, and calibrated action gates;
3. a falsifiable protocol separating retention, inference, allocation, functional drift, and calibration compensation;
4. an auditable implementation with claim-to-artifact traceability.

A successful thesis shows that at least one geometric object or constraint predicts and improves continual-system behaviour beyond strong non-geometric baselines under matched compute. A valuable negative result identifies where geometry fails and establishes simpler engineering rules.

The immediate next experiment is deliberately modest: instrument the current system for route regret, route–function alignment, calibration inflation, and saturation. Only when these diagnostics expose a real geometric control problem should G2 introduce projection, transport, Lie structure, or manifold-valued routing.

16 Conclusion

MMALS G2 reframes continual learning as the controlled historical evolution of routable functions. Chronicle supplies memory and provenance; inferred-context routing supplies conditional action; TPUT supplies utility and cost; CAL supplies calibrated uncertainty gates; geometry supplies neighbourhoods, directions, transport, and boundaries. The selected SFMDS paper contributes the strongest engineering pattern: expressive learning should occur inside an admissible solution space, with a clear distinction between soft encouragement and hard structural guarantees.

The research discipline remains conservative. No new mathematical layer enters the architecture until simpler diagnostics and controls fail. This keeps G2 a falsifiable engineering programme rather than a geometric metaphor.

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